12TH CHEMISTRY · CHAPTER 28 · FULL CHAPTER NOTES

Chromatography

Topic-wise short and long questions with answers from the complete chapter

28.1 INTRODUCTION

Q.1 [Short] What is chromatography? Which mixtures can be separated by it?

Separating one or two components from a simple mixture needs simple techniques such as filtration, distillation and solvent extraction; complex mixtures require advanced techniques. **Chromatography** is a versatile and important analytical technique that allows chemists to separate the individual components of a complex mixture based on their varying affinities for two different phases (a stationary phase and a mobile phase). Complex mixtures such as dyes in flowers, pollutants in a water body or medicinal extracts from plant leaves can be separated by it.

Q.2 [Short] Why is chromatography an important analytical tool? Name its three important types.

Chromatography allows the identification, purification and quantification of substances with high precision. This makes it an efficient modern analytical tool in medicine, forensic science, environmental science, etc. It also consumes only a very small quantity of the mixture. Three important types are paper chromatography (PC), thin layer chromatography (TLC) and gas chromatography (GC).

Q.3 [Short] What is adsorption? Give an example.

Adsorption is a surface phenomenon in which particles accumulate only on the outer surface of a material. For example, activated charcoal adsorbs gases and impurities onto its surface, which is why it is used in filters and purification systems.

Q.4 [Long] Explain the terms stationary phase, mobile phase, baseline, solvent front, R_f value and retention time.

Term	Meaning
Stationary phase	The phase that does not move: water trapped in the paper (PC), the solid layer of silica or alumina (TLC), or a non-volatile liquid on solid particles (GC).
Mobile phase	The phase that moves and carries the components: the solvent (PC and TLC) or an inert carrier gas (GC).
Baseline	The straight pencil line near the bottom of the paper or slide on which the spot of sample is placed.
Solvent front	The highest point reached by the solvent.
R_f value	Distance travelled by the substance divided by the distance travelled by the solvent.
Retention time	The time taken by a substance to pass through the column in GC.

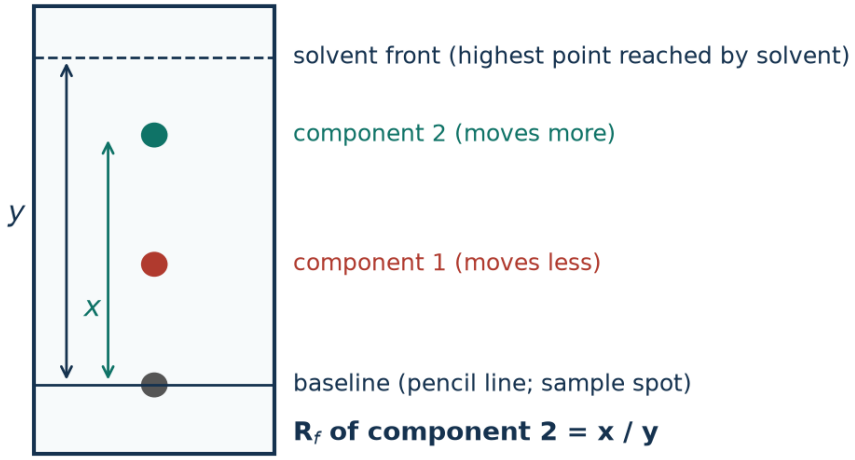
28.2 PAPER CHROMATOGRAPHY

Q.5 [Short] What is the principle of paper chromatography?

The technique of using a solvent to separate a mixture into its components on paper is called paper chromatography. The separating principle is the **relative solubility** (partition) of the components between two phases: the solvent moving up the paper (mobile phase) and the water combined with the cellulose of the paper (stationary phase).

Q.6 [Long] Describe the working of paper chromatography.

1. A small spot of the sample is put on the baseline near the bottom of the paper with a capillary tube.
2. The paper is dipped in a suitable solvent, which soaks up the paper. The solvent level must be below the spot of the sample.
3. The solutes in the sample dissolve in the solvent to different extents. Those more soluble in the solvent (mobile phase) move up the paper with it.
4. Others dissolve better in the water trapped in the paper (stationary phase), so they do not travel very far up the paper.
5. This difference in solubility separates the components. Colourless substances are made visible by spraying a locating agent which gives colour to them.
6. The R_f value of each spot is calculated and compared with known values.



Measuring the distances travelled by a spot and by the solvent front

Q.7 [Short] Why must the solvent level be below the spot of the sample?

If the spot were below the solvent level, the sample would dissolve directly into the solvent in the container instead of being carried up the paper, and no separation would take place.

Q.8 [Short] How are colourless substances made visible on a chromatogram?

They are sprayed with a locating agent which gives colour to them (e.g. ninhydrin for amino acids). In TLC, ultraviolet light or iodine vapours can also be used.

Q.9 [Long] What is the R_f value? How is it calculated and why is it useful?

The R_f value (retardation factor or retention factor) of a substance is calculated by dividing the distance travelled by the component by the distance travelled by the solvent.

Rf = distance travelled by the substance ÷ distance travelled by the solvent

R_f has no unit because it is a ratio. The R_f value of a substance does not change under the same conditions (same solvent and same temperature), so it is used to identify the substance.

Q.10 [Short] A spot travels 3.0 cm while the solvent front travels 7.5 cm. Calculate the R_f value.

Rf = 3.0 ÷ 7.5 = 0.40

28.3 THIN-LAYER CHROMATOGRAPHY

Q.11 [Short] How is a TLC plate prepared?

In thin layer chromatography (TLC), alumina (Al_2O_3) or silica (SiO_2) is mixed with water to make a slurry, which is spread evenly onto a glass slide. This layer is dried in an oven to make a white solid layer (the stationary phase) on the glass slide. TLC is faster than paper chromatography.

Q.12 [Long] Describe the working of thin layer chromatography.

1. A straight baseline is drawn with pencil and ruler near the bottom of the glass slide and a spot of sample is put on it with a capillary tube. More than one sample can be spotted at different places on the same slide.
2. The solvent is taken in a chromatography tank and the slide is dipped in it (mobile phase) so that the baseline is above the solvent level.
3. The components dissolve in the solvent and adsorb onto the aluminium oxide differently: some are more soluble in the mobile phase while some are more adsorbed onto the solid stationary phase.
4. After the solvent has moved up to a certain height, the slide is removed. The solvent front is marked and the slide is dried with a dryer.
5. The R_f values of the spots are calculated and compared with standard values to identify the components.

Q.13 [Short] Explain the principle of TLC. Why do polar substances travel slowly?

Principle (adsorption + solubility): polar substances have a greater attraction for the polar solid stationary phase and are adsorbed more strongly, so they travel more slowly up the plate. Non-polar substances are more soluble in the non-polar solvent and move further up the slide. This difference allows the substances in the sample to separate.

Q.14 [Short] Which locating agents are used in TLC?

When the components are colourless, a locating agent is used to visualise them. Ninhydrin spray gives a purple colour to colourless amino acids. Some colourless substances glow when irradiated with ultraviolet light, while some turn brown with iodine vapours.

Q.15 [Long] Give the applications of TLC, including its use in forensic chemistry.

1. It identifies unknown compounds by comparing their R_f values with standard known values.
2. The purity of a substance can be checked.
3. The progress of a reaction (whether reactants are consumed or products are made) can be checked.
4. In the pharmaceutical and food industry, the purity of drugs and the presence of preservatives, colourants and sweeteners in food can be checked.
5. It can detect pollutants, pesticides and toxic substances in soil, water, food and air.
6. In forensic science it is used to identify drugs, post-blast debris, blood, sweat and semen at a crime scene.

Q.16 [Long] What are the limitations of TLC?

1. It is mainly a qualitative technique and gives only an idea of the amount of a substance; it is not an accurate and precise quantitative technique.
2. It cannot clearly separate compounds having close R_f values, so the resolution of spots is not always high and sharp.
3. Only small amounts of sample can be applied and analysed; it is not suitable for large-scale separation.
4. It is sensitive to humidity, temperature and cleanliness of the working environment, which reduces reproducibility.

studysapiens.com

Q.17 [Short] **Why is the choice of stationary and mobile phases important in chromatography?**

Separation depends on the different affinities of the components for the two phases, so the phases must be chosen to suit the mixture. A polar stationary phase (silica or alumina) holds polar substances strongly, while a non-polar solvent carries non-polar substances further. If the solvent is too good for all components they all move with the solvent front; if it is too poor they all stay near the baseline, and in both cases the spots do not separate. A suitable pair of phases therefore gives well-separated spots (or peaks) with different R_f values (or retention times).

This learning outcome is not treated under a separate heading in the chapter; the answer applies the principles given for TLC.

Quick Check 28.1

Q.18 [Short] **a) A mixture contains butanone and pentane. How can they be separated by using thin layer chromatography using aluminium oxide as stationary phase and benzene as solvent?**

A spot of the mixture is placed on the baseline of the alumina plate and the plate is dipped in benzene (non-polar solvent) with the baseline above the solvent level. Butanone is polar (C=O group), so it is adsorbed more strongly on the polar aluminium oxide and travels slowly. Pentane is non-polar, is more soluble in benzene and moves further up the plate. The two therefore separate into different spots: pentane with the larger R_f value and butanone with the smaller one.

Q.19 [Short] **b) A mixture of two substances X and Y is analysed by thin-layer chromatography. The R_f value of substance X is larger than that of substance Y. Suggest why substance X has a larger R_f value.**

X is less polar than Y. It is adsorbed less strongly on the polar stationary phase and is more soluble in the solvent (mobile phase), so it travels further up the plate.

Q.20 [Short] **c) A separate thin layer chromatography experiment was done using the hair dyes A, B and C. The chromatogram obtained is shown. State three conclusions about the hair dyes A, B and C which can be deduced from the chromatogram. [A gives one low spot, B gives one high spot, C gives two spots at the same heights as those of A and B.]**

1. Dye A is a single substance (one spot), and so is dye B.
2. Dye C is a mixture of two substances (two spots).
3. Dye C contains both A and B, because its two spots are at the same heights (same R_f values) as the spots of A and B. (A and B are different substances.)

28.4 GAS CHROMATOGRAPHY

Q.21 [Short] What is gas chromatography? What are its stationary and mobile phases?

Gas chromatography (GC) is used to separate gases, volatile liquids and solids (in gaseous form). The **stationary phase** is a non-volatile, long-chain, non-polar hydrocarbon liquid of high boiling point, bonded to small particles of silica (SiO_2) or alumina (Al_2O_3). The **mobile phase** is an inert gas (carrier gas) such as helium or nitrogen.

Q.22 [Long] Describe the working of gas chromatography.

1. The sample is injected into the injector port, where it is quickly vaporised by an oven.
2. The carrier gas carries the vapours through a long coiled column packed with tiny silica particles of large surface area, over which separation occurs.
3. Non-polar substances interact more with the non-polar liquid of the stationary phase and are carried through the column more slowly. Substances that interact more strongly move slower while others move faster, so they separate.
4. As the separated substances leave the column they pass through a detector, which identifies and records them as peaks on a graph called a chromatogram. The detector also records the retention time of each component.



Main components of a gas chromatograph

Q.23 [Short] What is retention time?

The time taken by a substance to pass through the column is its retention time. Each component of a mixture has its own retention time under the same conditions, so the retention time is used to identify the component.

Q.24 [Long] How is the composition of a mixture found from a gas chromatogram?

GC is also a quantitative technique. The peaks are first identified and then the area of each peak is measured, which gives the proportion of each component in the mixture. The peaks are similar to triangles, so:

$$\text{Area of a peak} = \tfrac{1}{2} \times \text{base} \times \text{height}$$

$$\%\text{ of A} = (\text{peak area of A} \div \text{sum of peak areas of A, B and C}) \times 100$$

If the peaks are very narrow or have similar base widths, the peak height is used instead of the area to estimate the proportion of the components.

Q.25 [Short] A gas chromatogram shows peak areas of 20, 30 and 50 for X, Y and Z. Find the percentage of Y.

$$\% \text{ of Y} = (30 \div 100) \times 100 = 30\ \%$$

Q.26 [Short] What are the advantages of gas chromatography?

Gas chromatography has high sensitivity, good resolution, fast analysis and accurate quantification. A mixture of gases can easily be separated by GC at room temperature. It is used to check the presence of steroids in competing athletes and to test the components of the fuel used in car racing.

Q.27 [Long] Give the applications of gas chromatography.

1. **Industry:** vital for quality control and environmental protection; petrochemical plants use it to analyse the composition of fuels and to detect trace amounts of organic pollutants in environmental samples.
2. **Forensic laboratories:** provides evidence in legal investigations; used for blood alcohol content (BAC) tests and to detect illegal drugs, poisons or performance-enhancing substances in biological samples.
3. **Food and beverage industry:** ensures consistency and safety of products; detects spoilage by identifying metabolic by-products of bacteria or fungi before they are visible; analyses the aromatic profiles of perfumes, essential oils and food flavourings.
4. **Clinical and pharmaceutical:** diagnoses metabolic disorders by analysing specific compounds in a patient's breath or urine; tests the purity of newly synthesised drugs so that no harmful residual solvents or impurities remain.

Q.28 [Short] How did gas chromatography prove a man innocent?

Chromatography is important in forensic science. A man in China, Nian Bin, was accused of murdering two neighbouring children in 2005 by mixing rat poison (a fluoroacetate) in their food. After 8 years, in 2013, he was proven innocent and released: the food sample was tested using gas chromatography, which proved that no poison had been mixed.

Q.29 [Short] What are the limitations of gas chromatography?

1. GC works only for volatile (easily vaporised) substances.
2. Samples must be thermally stable; compounds that decompose at high temperatures cannot be used.
3. Large biomolecules like proteins and polymers cannot be analysed because they do not vaporise easily.
4. Instrument and maintenance costs are high.

studysapiens.com

Quick Check 28.2

Q.30 [Short] **An impure sample of lactic acid, $CH_3CH(OH)COOH$ contains butanone as the only contamination. The mixture is analysed using gas chromatography. Butanone is found to have a longer retention time than lactic acid. i) Suggest suitable substances that could be used as the mobile and stationary phases. ii) Describe how the percentage composition of the mixture can be determined from the gas chromatogram.**

i) Mobile phase: an inert carrier gas such as helium or nitrogen. Stationary phase: a non-volatile, long-chain, non-polar hydrocarbon liquid of high boiling point on silica (or alumina) particles. (Butanone is less polar than lactic acid, so it interacts more with this non-polar liquid and takes longer to pass through.)

ii) Identify the two peaks from their retention times. Measure the area of each peak (½ × base × height). Then:

$$\%\ \text{of lactic acid} = (\text{peak area of lactic acid} \div \text{sum of both peak areas}) \times 100$$

$$\%\ \text{of butanone} = (\text{peak area of butanone} \div \text{sum of both peak areas}) \times 100$$

28.5 COMPARISON OF THREE TYPES

Q.31 [Long] **Compare paper chromatography, TLC and gas chromatography.**

Feature	Paper chromatography	TLC (thin-layer)	GC (gas chromatography)
Principle	Partition	Adsorption + solubility	Partition (gas–liquid)
Mobile phase	Liquid	Liquid	Gas
Stationary phase	Water in paper	Silica/alumina	Non-volatile liquid
Speed	Slow	Faster	Very fast